

Math 571 - Homework 3

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Problems numbered [B:2.1:3] are from [Bergman's exercise notes](#). Problems labeled [R:2.1] are directly from Rudin. If I add an asterisk (*) to a problem, we have a slight variant of the problem.

Notation: $\{p_i\} = \{p_i \mid i = 1, 2, \dots\} \subseteq X$ is a set, which may or may not be infinite. $(p_i) = (p_i \mid i = 1, 2, \dots) = (p_1, p_2, \dots) \in X^{\mathbb{N}}$ is an infinite sequence from X . We might start counting from 0 instead of 1 on occasion, and if we need to start from another point, we make this explicit, using notation like $\{p_i\}_{i=k}^{\infty}$ and $(p_i)_{i=k}^{\infty}$. It is important to have these two notations, for example, if $p_i = (-1)^i$, then $\{p_i\} = \{-1, 1\}$ while $(p_i) = (-1, 1, -1, 1, \dots)$ and if $q_i = (-1)^{i+1}$, then $\{q_i\} = \{p_i\} = \{-1, 1\}$ while $(q_i) = (1, -1, 1, -1, \dots) = -(p_i)$.

Problem 3.1 (B:3.1:1 - A convergent sequence together with the point it approaches form a compact set.). Let X be a metric space.

- (a) Show that if (p_i) is a sequence from X with $p = \lim_{i \rightarrow \infty} p_i = p$, then $\{p\} \cup \{p_i\}$ is compact.
- (b) Deduce that for all $S \subseteq X$, S is closed iff for all compact $K \subseteq X$, $S \cap K$ is closed.

Problem 3.2 (B:3.2.1 - Converse to Theorem 3.6 (a).). Let X be a metric space. Show that if every sequence (x_i) in X has a convergent subsequence, then X is compact.

Hint: You can use R:2.26 which was homework problem 2.7.

Note: The statement: Every sequence has a convergent subsequence is called *sequential compactness*. So this problem has you show that if X is metric, then: X is sequentially compact iff X is compact.

Problem 3.3 (R:3.20). Let X be a metric space and (p_i) be a Cauchy sequence with a convergent subsequence. Show that (p_i) converges.

Problem 3.4 (R:3.22 - Baire Category Theorem). Suppose X is a complete metric space and $\{G_n\}$ is a set of dense open subsets. Show that $\bigcap G_n$ is dense.

Problem 3.5 (Application of Baire Category).

- (a) Show that $\mathbb{Q}$ is a countable union of closed sets.
- (b) Show that $\mathbb{Q}$ is not a countable intersection of open sets.

Problem 3.6 (R:3:16(18)*). Fix $\alpha > 1$ and $x_1 > \sqrt{\alpha}$, define

$$x_{n+1} = \frac{1}{2} \left(x_n + \frac{\alpha}{x_n} \right)$$

- a) Prove that (x_n) decreases monotonically and $\lim_{n \rightarrow \infty} x_n = \sqrt{\alpha}$.
- b) Let $\varepsilon_n = x_n - \sqrt{\alpha}$ be the error at the n^{th} term, show that

$$\varepsilon_{n+1} = \frac{\varepsilon_n^2}{2x_n} < \frac{\varepsilon_n^2}{2\sqrt{\alpha}}$$

Setting $\beta = 2\sqrt{\alpha}$, gives

$$\varepsilon_{n+1} < \beta \left(\frac{\varepsilon_1}{\beta} \right)^{2^n}$$

- c) Choose a number $\alpha > 3$ find a bound for how many terms are need to compute $\sqrt{\alpha}$ correct to 20 decimal places where x_1 is chosen minimally so that $x_1^2 > \alpha$. Prove your answer and do the computation. You might use [Python](#) or MATLAB.
- d) Replace the recursion above by

$$x_{n+1} = \frac{p-1}{p} x_n + \frac{\alpha}{p} x_n^{-p+1}$$

Discuss the behavior of (x_n) under suitable conditions. Don't bother trying to compute a recursive expression for ε_n in this case, but do prove your claims.

Problem 3.7 (B:3.7:1 - No boundary between convergent and divergent series.).

Problem 3.8 (R:3.9). Find the radius of convergence and **region interval of convergence** of the following power series. If you take $z \in \mathbb{R}$, then the region of convergence is an interval, but if $z \in \mathbb{C}$, then this region is the open disk $N_R(0)$ together with part of its boundary. The complex region might be difficult to calculate, but give it a try for the challenge. In all but one, the real interval is simple to calculate.

- a) $\sum_n n^3 z^n$
- b) $\sum_n \frac{2^n}{n!} z^n$
- c) $\sum_n \frac{2^n}{n^2} z^n$
- d) $\sum_n \frac{n^3}{3^n} z^n$
- e) $\sum_n \frac{n^n}{n!} x^n$

Problem 3.9 (B:3.13:1 - Radius of Convergence of sum and product series.). Let $\sum_n a_n$ and $\sum_n b_n$ be two series and let $\sum_n a_n + \sum_n b_n = \sum_n s_n$ and $(\sum_n a_n)(\sum_n b_n) = \sum_n p_n$. Recall that this means $s_n = a_n + b_n$ while $s_n = \sum_{i+j=n} a_i b_j = \sum_{i=1}^n a_i b_{n-i}$. Let R_a , R_b , R_s , and R_p be the radius of convergence of $\sum_n a_n$, $\sum_n b_n$, $\sum_n s_n$, and $\sum_n p_n$ respectively.

- (a) Show that if $\min\{R_a, R_b\} \leq \min\{R_s, R_p\}$, then

- (b) Deduce that if $R_a < R_b$, then $R_s = R_a$. What might happen if $R_a = R_b$?
- (c) Deduce that if $R_p < R_a$, then $R_b \leq R_p$.

Problem 3.10 (B:3.14:2 - Which series have convergent rearrangements?). Find a simple and easy-to-check criterion for a series $\sum a_n$ to have a rearrangement $\sum a_{\sigma(n)}$ that converges, for some σ a permutation of $\mathbb{Z}^+$.